Multi-Turn Speech Dialogue

A curated paper list for Multi-Turn Speech Dialogue.

Survey

1. "Turn-taking in Conversational Systems and Human-Robot Interaction: A Review". *Gabriel Skantze* Computer Speech & Language 2021. Paper
2. "A Review of Dialogue Systems: From Trained Monkeys to Stochastic Parrots". *Atharv Singh Patlan et al.* arXiv 2023. Paper
3. "WavChat: A Survey of Spoken Dialogue Models". *Shengpeng Ji et al.* arXiv 2024. Paper
4. "From Turn-Taking to Synchronous Dialogue: A Survey of Full-Duplex Spoken Language Models". *Yuxuan Chen and Haoyuan Yu* arXiv 2025. Paper
5. "A Survey of Recent Advances on Turn-taking Modeling in Spoken Dialogue Systems". *Galo Castillo-López et al.* IWSDS 2025. Paper
6. "Turn-Taking Modelling in Conversational Systems: A Review of Recent Advances". *Rutherford Agbeshi Patamia et al.* Technologies 2025. Paper
7. "A Survey of Full-Duplex Spoken Dialogue Systems: Architectural Hierarchy, Interaction Ontology, and Decision State Machine". *Jingyu Lu et al.* arXiv 2026. Paper

General Surveys

1. "A Survey of Available Corpora For Building Data-Driven Dialogue Systems: The Journal Version". *Iulian V. Serban et al.* Dialogue & Discourse 2018. Paper
2. "Neural Approaches to Conversational AI". *Jianfeng Gao et al.* Foundations and Trends in Information Retrieval 2019. Paper
3. "Recent Neural Methods on Dialogue State Tracking for Task-Oriented Dialogue Systems: A Survey". *Vevake Balaraman et al.* SIGDIAL 2021. Paper
4. "Who Says What to Whom: A Survey of Multi-Party Conversations". *Jia-Chen Gu et al.* IJCAI 2022. Paper
5. "Transformers in Speech Processing: A Survey". *Siddique Latif et al.* arXiv 2023. Paper
6. "A Survey on Recent Advances in LLM-Based Multi-turn Dialogue Systems". *Zihao Yi et al.* arXiv 2024. Paper
7. "Preference Tuning with Human Feedback on Language, Speech, and Vision Tasks: A Survey". *Genta Indra Winata et al.* arXiv 2024. Paper
8. "A Survey on Speech Large Language Models". *Jing Peng et al.* arXiv 2024. Paper
9. "On The Landscape of Spoken Language Models: A Comprehensive Survey". *Siddhant Arora et al.* arXiv 2025. Paper
10. "Recent Advances in Speech Language Models: A Survey". *Wenqian Cui et al.* ACL 2025. Paper
11. "Aligning Multimodal LLM with Human Preference: A Survey". *Tao Yu et al.* arXiv 2025. Paper

Datasets

1. "Japanese Dialogue Corpus of Information Navigation and Attentive Listening Annotated with Extended ISO-24617-2 Dialogue Act Tags". *Koichiro Yoshino et al.*. LREC 2018. Paper
2. "Taskmaster-1: Toward a Realistic and Diverse Dialog Dataset". *Bill Byrne et al.*. EMNLP-IJCNLP 2019. Paper
3. "Alexa in the Wild: Collecting Unconstrained Conversations with a Modern Voice Assistant in a Public Environment". *Benjamin Benk et al.*. LREC 2020. Paper
4. "HarperValleyBank: A Domain-Specific Spoken Dialog Corpus". *Mike Wu et al.*. arXiv 2020. Paper
5. "The MSP-Conversation Corpus". *Luz Martinez-Lucas et al.*. Interspeech 2020. Paper
6. "ASCEND: A Spontaneous Chinese-English Dataset for Code-switching in Multi-turn Conversation". *Holy Lovenia et al.*. arXiv 2021. Paper
7. "DailyTalk: Spoken Dialogue Dataset for Conversational Text-to-Speech". *Keon Lee et al.* arXiv 2022. Paper
8. "DOROTHIE: Spoken Dialogue for Handling Unexpected Situations in Interactive Autonomous Driving Agents". *Ziqiao Ma et al.* EMNLP Findings 2022. Paper
9. "Collection and Analysis of Travel Agency Task Dialogues with Age-Diverse Speakers". *Michimasa Inaba et al.*. LREC 2022. Paper
10. "Design and Evaluation of the Corpus of Everyday Japanese Conversation". *Hanae Koiso et al.*. LREC 2022. Paper
11. "SpokenWOZ: A Large-Scale Speech-Text Benchmark for Spoken Task-Oriented Dialogue Agents". *Shuzheng Si et al.* arXiv 2023. Paper
12. "CALLS: Japanese Empathetic Dialogue Speech Corpus of Complaint Handling and Attentive Listening in Customer Center". *Yuki Saito et al.* Interspeech 2023. Paper
13. "Advancing Large Language Models to Capture Varied Speaking Styles and Respond Properly in Spoken Conversations (StyleTalk)". *Guan-Ting Lin et al.* ACL 2024. Paper
14. "Generative Expressive Conversational Speech Synthesis (NCSSD)". *Rui Liu et al.* ACM MM 2024. Paper
15. "J-CHAT: Japanese Large-scale Spoken Dialogue Corpus for Spoken Dialogue Language Modeling". *Wataru Nakata et al.* arXiv 2024. Paper
16. "DeepDialogue: A Multi-Turn Emotionally-Rich Spoken Dialogue Dataset". *Alkis Koudounas et al.* arXiv 2025. Paper
17. "UltraVoice: Scaling Fine-Grained Style-Controlled Speech Conversations for Spoken Dialogue Models". *Wenming Tu et al.* arXiv 2025. Paper
18. "Toward Conversational Hungarian Speech Recognition: Introducing the BEA-Large and BEA-Dialogue Datasets". *Mate Gedeon et al.* arXiv 2025. Paper
19. "InteractSpeech: A Speech Dialogue Interaction Corpus for Spoken Dialogue Model". *Yifu Chen et al.* EMNLP Findings 2025. Paper
20. "Behavior-SD: Behaviorally Aware Spoken Dialogue Generation with Large Language Models". *Sehun Lee et al.* NAACL 2025. Paper

21. "RealTalk-CN: A Realistic Chinese Speech-Text Dialogue Benchmark With Cross-Modal Interaction Analysis". *Enzhi Wang et al.* arXiv 2025. Paper
22. "DialogueAgents: A Hybrid Agent-Based Speech Synthesis Framework for Multi-Party Dialogue". *Zheyuan Zhang et al.* arXiv 2025. Paper
23. "Open-Source Full-Duplex Conversational Datasets for Natural and Interactive Speech Synthesis". *Rui Liu et al.* arXiv 2025. Paper
24. "Exploring Emotional Nuances in Spoken Dialogue: Dataset Construction and Prediction of Emotional Dialogue Breakdown". *Hyuga Nakaguro et al.* IWSDS 2026. Paper
25. "DuplexChat: Constructing Speaker-Separated Full-Duplex Dialogue Speech at Scale for Spoken Dialogue Language Modeling". *Wataru Nakata et al.* arXiv 2026. Paper
26. "Dial HEALTHDIAL for Advice: A Multilingual and Multi-Parallel Spoken Dialogue Dataset for Knowledge-Grounded Information Seeking". *Songbo Hu et al.* ACL Findings 2026. Paper

Evaluation & Benchmarks

Evaluation Focus: Turn-taking & Interruption

1. "Investigating Speech Features for Continuous Turn-Taking Prediction Using LSTMs". *Matthew Roddy et al..* Interspeech 2018. Paper
2. "Turn-Taking Predictions Across Languages and Genres Using an LSTM Recurrent Neural Network". *Nigel G. Ward et al..* SLT 2018. Paper
3. "How Much Does Prosody Help Turn-taking? Investigations using Voice Activity Projection Models". *Erik Ekstedt et al.* SIGDIAL 2022. Paper
4. "Talking Turns: Benchmarking Audio Foundation Models on Turn-Taking Dynamics". *Siddhant Arora et al.* ICLR 2025. Paper
5. "Investigating the Impact of Incremental Processing and Voice Activity Projection on Spoken Dialogue Systems". *Yuya Chiba et al.* COLING 2025. Paper
6. "Synchronization and Turn-Taking in Full-Duplex Speech Dialogue Models". *Pablo Riera et al.* arXiv 2026. Paper
7. "Still Between Us? Evaluating and Improving Voice Assistant Robustness to Third-Party Interruptions". *Dongwook Lee et al.* ACL 2026. Paper
8. "Rethinking Binary Evaluation of Turn-Taking under Inherent Ambiguity". *Yunosuke Kubo et al.* SIGDIAL 2026. Paper

Evaluation Focus: Full-duplex Interaction

1. "MMedFD: A Real-world Healthcare Benchmark for Multi-turn Full-Duplex Automatic Speech Recognition". *Hongzhao Chen et al.* arXiv 2025. Paper
2. "Full-Duplex-Bench: A Benchmark to Evaluate Full-duplex Spoken Dialogue Models on Turn-taking Capabilities". *Guan-Ting Lin et al.* arXiv 2025. Paper
3. "FD-Bench: A Full-Duplex Benchmarking Pipeline Designed for Full Duplex Spoken Dialogue Systems". *Yizhou Peng et al.* Interspeech 2025. Paper

4. "The ICASSP 2026 HumDial Challenge: Benchmarking Human-like Spoken Dialogue Systems in the LLM Era". *Zhixian Zhao et al.* ICASSP / arXiv 2026. Paper
5. "τ-Voice: Benchmarking Full-Duplex Voice Agents on Real-World Domains". *Soham Ray et al.* ICML 2026. Paper
6. "Full-Duplex-Bench-v3: Benchmarking Tool Use for Full-Duplex Voice Agents Under Real-World Disfluency". *Guan-Ting Lin et al.* arXiv 2026. Paper
7. "Full-Duplex Interaction in Spoken Dialogue Systems: A Comprehensive Study from the ICASSP 2026 HumDial Challenge". *Chengyou Wang et al.* arXiv 2026. Paper
8. "Full-Duplex-Bench-v2: A Multi-Turn Evaluation Framework for Duplex Dialogue Systems with an Automated Examiner". *Guan-Ting Lin et al.* ACL 2026. Paper
9. "MTR-DuplexBench: Towards a Comprehensive Evaluation of Multi-Round Conversations for Full-Duplex Speech Language Models". *Zhang He et al.* ACL Findings 2026. Paper
10. "M3-DuplexBench: A Multi-Turn, Multilingual, Multidomain Benchmark for Full-Duplex Spoken Dialogue Models". *Ryo Fukuda et al.* arXiv 2026. Paper

Evaluation Focus: Multi-turn Dialogue Capabilities

1. "SD-Eval: A Benchmark Dataset for Spoken Dialogue Understanding Beyond Words". *Junyi Ao et al.* NeurIPS 2024 (Datasets & Benchmarks). Paper
2. "Contextual Interactive Evaluation of TTS Models in Dialogue Systems". *Siyang Wang et al.* Interspeech 2024. Paper
3. "Benchmarking Open-ended Audio Dialogue Understanding for Large Audio-Language Models". *Kuofeng Gao et al.* arXiv 2024. Paper
4. "URO-Bench: Towards Comprehensive Evaluation for End-to-End Spoken Dialogue Models". *Ruiqi Yan et al.* EMNLP Findings 2025. Paper
5. "WavReward: Spoken Dialogue Models With Generalist Reward Evaluators". *Shengpeng Ji et al.* arXiv 2025. Paper
6. "MAD: A Benchmark for Multi-Turn Audio Dialogue Fact-Checking". *Chaewan Chun et al.* arXiv 2025. Paper
7. "VoxRole: A Comprehensive Benchmark for Evaluating Speech-Based Role-Playing Agents". *Weihao Wu et al.* arXiv 2025. Paper
8. "Evaluating Bias in Spoken Dialogue LLMs for Real-World Decisions and Recommendations (FairDialogue)". *Yihao Wu et al.* arXiv 2025. Paper
9. "MULTI-Bench: A Multi-turn Interactive Benchmark for Assessing Emotional Intelligence Ability of Spoken Dialogue Models". *Yayue Deng et al.* arXiv 2025. Paper
10. "Audio MultiChallenge: A Multi-Turn Evaluation of Spoken Dialogue Systems on Natural Human Interaction". *Advait Gosai et al.* arXiv 2025. Paper
11. "VoxDialogue: Can Spoken Dialogue Systems Understand Information Beyond Words?". *Xize Cheng et al.* ICLR 2025. Paper
12. "EchoMind: An Interrelated Multi-level Benchmark for Evaluating Empathetic Speech Language Models". *Li Zhou et al* arXiv 2025. Paper
13. "SOVA-Bench: Benchmarking the Speech Conversation Ability for LLM-based Voice Assistant". *Yixuan Hou et al.* Interspeech 2025. Paper

14. "MTalk-Bench: Evaluating Speech-to-Speech Models in Multi-Turn Dialogues via Arena-style and Rubrics Protocols". *Yuhao Du et al.* arXiv 2025. Paper
15. "VoiceAgentBench: Are Voice Assistants Ready for Agentic Tasks?". *Dhruv Jain et al.* arXiv 2025. Paper
16. "C3: A Bilingual Benchmark for Spoken Dialogue Models Exploring Challenges in Complex Conversations". *Chengqian Ma et al.* EMNLP 2025. Paper
17. "HumDial-EIBench: A Human-Recorded Multi-Turn Emotional Intelligence Benchmark for Audio Language Models". *Shuiyuan Wang et al* arXiv 2026. Paper
18. "VoiceBench: Benchmarking LLM-Based Voice Assistants". *Yiming Chen et al.* TACL 2026. Paper
19. "The Context Trap: Why End-to-End Audio Language Models Fail Multi-turn Dialogues". *Zhi Rui Tam et al.* IWSDS 2026. Paper
20. "SDiaReward: Modeling and Benchmarking Spoken Dialogue Rewards with Modality and Colloquialness". *Jingyu Lu et al.* ACL 2026. Paper
21. "SpeakerSleuth: Can Large Audio-Language Models Judge Speaker Consistency across Multi-turn Dialogues?". *Jonggeun Lee et al.* ACL 2026. Paper
22. "Style Amnesia: Investigating Speaking Style Degradation and Mitigation in Multi-Turn Spoken Language Models". *Yu-Xiang Lin et al.* ACL Findings 2026. Paper

Models

Interaction Mode: Turn-taking

1. "Improving End-of-Turn Detection in Spoken Dialogues by Detecting Speaker Intentions as a Secondary Task". *Zakaria Aldeneh et al..* ICASSP 2018. Paper
2. "Neural Dialogue Context Online End-of-Turn Detection". *Ryo Masumura et al..* SIGDIAL 2018. Paper
3. "Prediction of Turn-Taking Using Multitask Learning with Prediction of Backchannels and Fillers". *Kohei Hara et al..* Interspeech 2018. Paper
4. "Turn-Taking Prediction Based on Detection of Transition Relevance Place". *Kohei Hara et al..* Interspeech 2019. Paper
5. "TurnGPT: A Transformer-Based Language Model for Predicting Turn-Taking in Spoken Dialog". *Erik Ekstedt and Gabriel Skantze.* EMNLP Findings 2020. Paper
6. "Projection of Turn Completion in Incremental Spoken Dialogue Systems". *Erik Ekstedt and Gabriel Skantze.* SIGDIAL 2021. Paper
7. "Timing Generating Networks: Neural Network Based Precise Turn-Taking Timing Prediction in Multiparty Conversation". *Shinya Fujie et al..* Interspeech 2021. Paper
8. "Using Transition Duration to Improve Turn-taking in Conversational Agents". *Charles Threlkeld et al.* SIGDIAL 2022. Paper
9. "When can I Speak? Predicting Initiation Points for Spoken Dialogue Agents". *Siyan Li et al.* SIGDIAL 2022. Paper

10. "Voice Activity Projection: Self-supervised Learning of Turn-taking Events". *Erik Ekstedt and Gabriel Skantze* Interspeech 2022. Paper
11. "Gated Multimodal Fusion with Contrastive Learning for Turn-Taking Prediction in Human-Robot Dialogue". *Jiudong Yang et al..* ICASSP 2022. Paper
12. "Response Timing Estimation for Spoken Dialog System Using Dialog Act Estimation". *Jin Sakuma et al..* Interspeech 2022. Paper
13. "Turn-Taking Prediction for Natural Conversational Speech". *Shuo-Yiin Chang et al..* Interspeech 2022. Paper
14. "Generative Spoken Dialogue Language Modeling". *Tu Anh Nguyen et al.* arXiv 2022 / TACL 2023. Paper
15. "SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities". *Dong Zhang et al.* EMNLP Findings 2023. Paper
16. "Let's Go Real Talk: Spoken Dialogue Model for Face-to-Face Conversation". *Se Park et al.* ACL 2024. Paper
17. "Qwen2-Audio Technical Report". *Yunfei Chu et al.* arXiv 2024. Paper
18. "Style-Talker: Finetuning Audio Language Model and Style-Based Text-to-Speech Model for Fast Spoken Dialogue Generation". *Yinghao Aaron Li et al.* arXiv 2024. Paper
19. "EMOVA: Empowering Language Models to See, Hear and Speak with Vivid Emotions". *Kai Chen et al.* arXiv 2024. Paper
20. "Internalizing ASR with Implicit Chain of Thought for Efficient Speech-to-Speech Conversational LLM". *Robin Shinghei Yuen et al.* NeurIPS 2024. Paper
21. "Building a Taiwanese Mandarin Spoken Language Model: A First Attempt". *Chih-Kai Yang et al.* arXiv 2024. Paper
22. "LLaMA-Omni: Seamless Speech Interaction with Large Language Models". *Qingkai Fang et al.* arXiv 2024. Paper
23. "Mini-Omni: Language Models Can Hear, Talk While Thinking in Streaming". *Zhifei Xie et al.* arXiv 2024. Paper
24. "GLM-4-Voice: Towards Intelligent and Human-Like End-to-End Spoken Chatbot". *Aohan Zeng et al.* arXiv 2024. Paper
25. "Triadic Multi-party Voice Activity Projection for Turn-taking in Spoken Dialogue Systems". *Mikey Elmers et al.* Interspeech 2025. Paper
26. "OpenOmni: Advancing Open-Source Omnimodal Large Language Models with Progressive Multimodal Alignment and Real-time Emotional Speech Synthesis". *Run Luo et al.* arXiv 2025. Paper
27. "GOAT-SLM: A Spoken Language Model with Paralinguistic and Speaker Characteristic Awareness". *Hongjie Chen et al.* arXiv 2025. Paper
28. "OpenS2S: Advancing Fully Open-Source End-to-End Empathetic Large Speech Language Model". *Chen Wang et al.* arXiv 2025. Paper
29. "Step-Audio 2 Technical Report". *StepFun Team et al.* arXiv 2025. Paper
30. "InteractiveOmni: A Unified Omni-modal Model for Audio-Visual Multi-turn Dialogue". *Wenwen Tong et al.* arXiv 2025. Paper

31. "Step-Audio: Unified Understanding and Generation in Intelligent Speech Interaction". *Ailin Huang et al.* arXiv 2025. [Paper]
32. "Baichuan-Audio: A Unified Framework for End-to-End Speech Interaction". *Tianpeng Li et al.* arXiv 2025. [Paper]
33. "Kimi-Audio Technical Report". *KimiTeam et al.* arXiv 2025. [Paper]
34. "Qwen3-Omni Technical Report". *Jin Xu et al.* arXiv 2025. [Paper]
35. "PACHAT: Persona-Aware Speech Assistant for Multi-party Dialogue". *Dongjie Fu et al.* EMNLP 2025. [Paper]
36. "VITA-Audio: Fast Interleaved Cross-Modal Token Generation for Efficient Large Speech-Language Model". *Zuwei Long et al.* arXiv 2025. [Paper]
37. "AV-Dialog: Spoken Dialogue Models with Audio-Visual Input". *Tuochao Chen et al.* arXiv 2025. [Paper]
38. "Qwen2.5-Omni Technical Report". *Qwen Team et al.* arXiv 2025. [Paper]
39. "LLaMA-Omni 2: LLM-based Real-time Spoken Chatbot with Autoregressive Streaming Speech Synthesis". *Qingkai Fang et al.* ACL 2025. [Paper]
40. "VoxMind: An End-to-End Agentic Spoken Dialogue System". *Tianle Liang et al.* ACL 2026. [Paper]
41. "ZipVoice-Dialog: Non-Autoregressive Spoken Dialogue Generation with Flow Matching". *Han Zhu et al.* ACL Findings 2026. [Paper]

Interaction Mode: Half-duplex / Controlled Barge-in

1. "An Incremental Turn-Taking Model for Task-Oriented Dialog Systems". *Andrei C. Coman et al..* Interspeech 2019. [Paper]
2. "Oh, Jeez! or Uh-Huh? A Listener-Aware Backchannel Predictor on ASR Transcriptions". *Daniel Ortega et al..* ICASSP 2020. [Paper]
3. "BPM_MT: Enhanced Backchannel Prediction Model Using Multi-Task Learning". *Jin Yea Jang et al..* EMNLP 2021. [Paper]
4. "Device Directedness with Contextual Cues for Spoken Dialog Systems". *Dhanush Bekal et al.* arXiv 2022. [Paper]
5. "Contextual Acoustic Barge-In Classification for Spoken Dialog Systems". *Dhanush Bekal et al..* Interspeech 2022. [Paper]
6. "Freeze-Omni: A Smart and Low Latency Speech-to-Speech Dialogue Model with Frozen LLM". *Xiong Wang et al.* arXiv 2024. [Paper]
7. "Mini-Omni2: Towards Open-source GPT-4o with Vision, Speech and Duplex Capabilities". *Zhifei Xie et al.* arXiv 2024. [Paper]
8. "VITA: Towards Open-Source Interactive Omni Multimodal LLM". *Chaoyou Fu et al.* arXiv 2024. [Paper]
9. "SHANKS: Simultaneous Hearing and Thinking for Spoken Language Models". *Cheng-Han Chiang et al.* arXiv 2025. [Paper]

Interaction Mode: Full-duplex

1. "Duplex Conversation in Outbound Agent System". *Chunxiang Jin et al.* Interspeech 2021. [Paper]

2. "Duplex Conversation: Towards Human-like Interaction in Spoken Dialogue Systems". *Ting-En Lin et al.* arXiv 2022. Paper
3. "Beyond Turn-Based Interfaces: Synchronous LLMs as Full-Duplex Dialogue Agents". *Bandhav Veluri et al.* EMNLP 2024. Paper
4. "Moshi: A Speech-Text Foundation Model for Real-Time Dialogue". *Alexandre Defossez et al.* arXiv 2024. Paper
5. "SALMONN-omni: A Codec-free LLM for Full-duplex Speech Understanding and Generation". *Wenyi Yu et al.* arXiv 2024. Paper
6. "A Full-duplex Speech Dialogue Scheme Based On Large Language Models". *Peng Wang et al.* NeurIPS 2024. Paper
7. "MinMo: A Multimodal Large Language Model for Seamless Voice Interaction". *FunAudioLLM Team et al.* arXiv 2025. Paper
8. "SALM-Duplex: Efficient and Direct Duplex Modeling for Speech-to-Speech Language Model". *Ke Hu et al.* arXiv 2025. Paper
9. "OmniFlatten: An End-to-end GPT Model for Seamless Voice Conversation". *Qinglin Zhang et al.* ACL 2025. Paper
10. "Towards a Japanese Full-duplex Spoken Dialogue System". *Atsumoto Ohashi et al.* Interspeech 2025. Paper
11. "FireRedChat: A Pluggable, Full-Duplex Voice Interaction System with Cascaded and Semi-Cascaded Implementations". *Junjie Chen et al.* arXiv 2025. Paper
12. "Incorporating Dialogue State Tracking into Japanese Full-duplex Task-oriented Spoken Dialogue Model". *Yuya Chiba et al.* IJCNLP-AACL 2025. Paper
13. "Chronological Thinking in Full-Duplex Spoken Dialogue Language Models". *Donghang Wu et al.* arXiv 2025. Paper
14. "LLM-Enhanced Dialogue Management for Full-Duplex Spoken Dialogue Systems". *Hao Zhang et al.* arXiv 2025. Paper
15. "Language Model Can Listen While Speaking". *Ziyang Ma et al.* AAAI 2025. Paper
16. "FlashLabs Chroma 1.0: A Real-Time End-to-End Spoken Dialogue Model with Personalized Voice Cloning". *Tanyu Chen et al.* arXiv 2026. Paper
17. "PersonaPlex: Voice and Role Control for Full Duplex Conversational Speech Models". *Rajarshi Roy et al.* arXiv 2026. Paper
18. "MiniCPM-o 4.5: Towards Real-Time Full-Duplex Omni-Modal Interaction". *Junbo Cui et al.* arXiv 2026. Paper
19. "DuplexSLA: A Full-Duplex Spoken Language Model with Synchronized Speech, Language, and Action". *Haoyang Zhang et al.* arXiv 2026. Paper
20. "BayLing-Duplex: Native Full-Duplex Speech Dialogue with a Single Autoregressive LLM". *Qingkai Fang et al.* arXiv 2026. Paper
21. "F-Actor: Controllable Conversational Behavior in Full-Duplex Models". *Maike Züfle et al.* ACL Findings 2026. Paper
22. "Hierarchical Acoustic-Semantic Modeling: Modality Separation and Semantic Coherence for Full-Duplex SLMs". *Zhenyu Liu et al.* ACL 2026. Paper

23. "JoyAI-Talker: Full-Duplex Speech Interactive Large Model Built for Empathetic Voice Agents". *Yinhao Bai et al.* arXiv 2026. [Paper]
24. "SoulX-Duplug: Plug-and-Play Streaming State Prediction Module for Realtime Full-Duplex Speech Conversation". *Ruiqi Yan et al.* arXiv 2026. [Paper]

Training Methods

Interaction Mode: Turn-taking

1. "Towards Generalized Models for Task-Oriented Dialogue Modeling on Spoken Conversations". *Ruijie Yan et al.*. arXiv 2022. [Paper]
2. "Adapting Text-based Dialogue State Tracker for Spoken Dialogues". *Jaeseok Yoon et al.* DSTC 2023. [Paper]
3. "SpeechAlign: Aligning Speech Generation to Human Preferences". *Dong Zhang et al.* NeurIPS 2024. [Paper]
4. "Data-Centric Improvements for Enhancing Multi-Modal Understanding in Spoken Conversation Modeling (ASK-QA)". *Maximillian Chen et al.* ACL Findings 2025. [Paper]
5. "WavRAG: Audio-Integrated Retrieval Augmented Generation for Spoken Dialogue Models". *Yifu Chen et al.* ACL 2025. [Paper]
6. "Enhancing Speech-to-Speech Dialogue Modeling with End-to-End Retrieval-Augmented Generation". *Pengchao Feng et al.* EMNLP Findings 2025. [Paper]
7. "Aligning Spoken Dialogue Models from User Interactions". *Anne Wu et al.* arXiv 2025. [Paper]
8. "Chain-of-Thought Training for Open E2E Spoken Dialogue Systems". *Siddhant Arora et al.* arXiv 2025. [Paper]
9. "VoiceTextBlender: Augmenting Large Language Models with Speech Capabilities via Single-Stage Joint Speech-Text Supervised Fine-Tuning". *Yifan Peng et al.* NAACL 2025. [Paper]
10. "Approaching Dialogue State Tracking via Aligning Speech Encoders and LLMs". *Šimon Sedláček et al.* arXiv 2025. [Paper]
11. "The Speech-LLM Takes It All: A Truly Fully End-to-End Spoken Dialogue State Tracking Approach". *Nizar El Ghazal et al.* arXiv 2025. [Paper]
12. "Optimizing Conversational Quality in Spoken Dialogue Systems with Reinforcement Learning from AI Feedback". *Siddhant Arora et al.* arXiv 2026. [Paper]
13. "WavAlign: Enhancing Intelligence and Expressiveness in Spoken Dialogue Models via Adaptive Hybrid Post-Training". *Yifu Chen et al.* ACL Findings 2026. [Paper]

Interaction Mode: Half-duplex / Streaming

1. "A Methodology for Turn-Taking Capabilities Enhancement in Spoken Dialogue Systems Using Reinforcement Learning". *Hatim Khouzaimi et al.*. Computer Speech & Language 2018. [Paper]
2. "Stream RAG: Instant and Accurate Spoken Dialogue Systems with Streaming Tool Usage". *Siddhant Arora et al.* arXiv 2025. [Paper]
3. "Dual-Reasoner: Bridging Interleaved Atomicity and Streaming Latency via Thinking-while-Talking". *Yangzhuo Li et al.* ACL Findings 2026. [Paper]

Interaction Mode: Full-duplex

1. "FLM-Audio: Natural Monologues Improves Native Full-Duplex Chatbots via Dual Training". *Yiqun Yao et al.* arXiv 2025. Paper
2. "Reinforcement Learning Enhanced Full-Duplex Spoken Dialogue Language Models for Conversational Interactions". *Chen Chen et al.* COLM 2025. Paper
3. "Multi-Faceted Interactivity Alignment in Full-Duplex Speech Models". *Atsumoto Ohashi et al.* arXiv 2026. Paper
4. "Reproducing Proficiency-Conditioned Dialogue Features with Full-duplex Spoken Dialogue Models". *Takao Obi et al.* IWSDS 2026. Paper
5. "Effects of Dialogue Corpora Properties on Fine-Tuning a Moshi-Based Spoken Dialogue Model". *Yuto Abe et al.* IWSDS 2026. Paper
6. "Dual-Axis Generative Reward Model Toward Semantic and Turn-taking Robustness in Interactive Spoken Dialogue Models". *Yifu Chen et al.* ACL 2026. Paper
7. "Sommelier: Scalable Open Multi-turn Audio Pre-processing for Full-duplex Speech Language Models". *Kyudan Jung et al.* ACL Industry Track 2026. Paper

Survey Coverage Comparison

Table 1. Coverage comparison of surveys related to Multi-Turn Speech Dialogue, including fine-grained organization aligned with this collection.

General Surveys are listed first, followed by core surveys. ✓ = systematic coverage; △ = partial or supporting coverage; - = not a substantive focus. Interaction Taxonomy denotes an explicit taxonomy of conversational interaction. Fine-grained Organization assesses whether a survey structures its analysis using the evaluation focuses, model interaction modes, and training interaction modes adopted in this collection; relevant discussion without a matching subdivision is marked △.

Survey	Research Scope			Content Coverage				Interaction Taxonomy	Fine-grained Organization		
	Multi-turn Speech	Turn-taking & Interruption	Full-duplex	Datasets	Evaluation / Benchmark	Models	Training Methods		Evaluation Focus	Model Interaction Modes	Training Interaction Modes
Serban et al. (2018)	△	-	-	✓	✓	△	△	✓	-	-	-
Gao et al. (2019)	△	-	-	△	△	✓	✓	✓	-	-	-
Balaraman et al. (2021)	△	-	-	✓	✓	✓	△	✓	△	-	-
Gu et al. (2022)	△	△	-	△	△	✓	△	✓	△	△	-
Latif et al. (2023)	△	-	-	△	△	✓	△	✓	-	-	-
Yi et al. (2024)	△	-	-	✓	✓	✓	✓	△	△	-	△
Winata et al. (2024)	-	-	-	✓	✓	△	✓	✓	-	-	△
Peng et al. (2024)	△	-	-	△	✓	✓	✓	✓	△	△	△
Arora et al. (2025)	△	△	△	△	✓	✓	✓	✓	△	△	△
Cui et al. (2025)	△	-	△	△	✓	✓	✓	✓	△	△	△
Yu et al. (2025)	-	-	-	✓	✓	△	✓	✓	-	-	△
Skantze (2021)	✓	✓	△	△	△	✓	-	✓	✓	✓	△
Patlan et al. (2023)	✓	△	-	△	✓	✓	✓	✓	△	△	△
Ji et al. / WavChat (2024)	✓	✓	✓	✓	✓	✓	✓	✓	△	△	△
Chen and Yu (2025)	✓	✓	✓	△	✓	✓	△	✓	✓	✓	△
Castillo-López et al. (2025)	✓	✓	-	✓	△	✓	△	✓	✓	✓	✓
Patamia et al. (2025)	✓	✓	△	✓	✓	✓	△	✓	✓	✓	✓
Lu et al. (2026)	✓	✓	✓	✓	✓	✓	△	✓	✓	✓	△
Current Collection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

System and Model Comparison

Table 2. Comparison of all papers listed under Models for Multi-Turn Speech Dialogue.

✓ = explicitly reported support; △ = partial, asymmetric, or controller-mediated support; - = not reported, not applicable, or not a primary capability. Method names link to the papers.

Method	Scope	Interaction Capabilities				Modalities		System Architecture			Speech Interface	Acoustic / Speech Model	Release
		Streaming	Barge-in	Concurrent	Backchannel / Overlap	Speech Output	Vision	Cascaded	Semi-casc.	End-to-end			
Interaction Mode: Turn-taking													
EOT + Intent	Component	△	-	-	-	-	-	-	-	-	Acoustic features	BiLSTM acoustic EOT/intent classifier	2018.04
Neural Online EOT	Component	✓	-	-	-	-	-	-	-	-	Acoustic + context	LSTM online EOT predictor (no generator)	2018.07
MTL Turn-taking	Component	△	-	-	✓	-	-	-	-	-	Acoustic features	Multitask LSTM turn/BC/filler predictor	2018.09
TRP Turn-taking	Component	△	-	-	-	-	-	-	-	-	Speech features	LSTM TRP predictor (no generator)	2019.09
TurnGPT	Component	△	-	-	-	-	-	-	-	-	Text tokens	GPT-2 turn-shift LM (text-only)	2020.11
Turn Completion Projection	Component	✓	-	-	-	-	-	-	-	-	Incremental text	Incremental completion predictor (no generator)	2021.07
Timing Generating Networks	Component	△	-	-	-	-	-	-	-	-	Timing features	LSTM timing-generating network	2021.08
Transition Duration	Component	△	-	-	-	-	-	-	-	-	Timing features	Transition-duration predictor (no generator)	2022.09
Initiation Point Predictor	Component	△	-	-	-	-	-	-	-	-	Speech + context	Neural initiation-point predictor (no generator)	2022.09
VAP	Component	✓	-	-	✓	-	-	-	-	-	Waveform	Causal Transformer voice-activity predictor	2022.09
Gated Multimodal Fusion	Component	△	-	-	-	-	✓	-	-	-	Audio-visual features	Gated audio-visual turn predictor	2022.05
Response Timing Estimator	Component	△	-	-	-	-	-	-	-	-	Dialogue acts	DNN response-timing estimator	2022.09
Natural TT Predictor	Component	△	-	-	-	-	-	-	-	-	Conversational speech	Neural turn-taking predictor (no generator)	2022.09
dGSLM	Model	△	-	△	✓	✓	-	-	-	✓	HuBERT units	HuBERT units + dual-stream Transformer + HiFi-GAN	2022.03
SpeechGPT	Model	-	-	-	-	✓	-	-	-	✓	HuBERT units	HuBERT units + LLaMA + Unit HiFi-GAN	2023.05
RealTalk	System	△	-	-	-	✓	-	-	-	✓	Speech features	SSL speech encoder + dialogue Transformer + unit decoder	2024.08
Qwen2-Audio	Model	-	-	-	-	-	-	-	✓	-	Whisper features	Whisper-large-v3 encoder + Qwen-7B	2024.07
Style-Talker	System	△	-	-	-	✓	-	-	✓	-	Audio LM + TTS	Speech encoder + audio LM + style-conditioned TTS	2024.08
EMOVA	Model	-	-	-	-	✓	✓	-	-	✓	Semantic + acoustic	Semantic-acoustic tokenizer + LLM + style decoder	2024.09
IntrinsicVoice	Model	△	-	-	-	✓	-	-	-	✓	HuBERT units	HuBERT units + LLaMA + GroupFormer decoder	2024.09
Taiwanese Mandarin SLM	Model	-	-	-	-	✓	-	-	-	✓	Discrete units	Discrete-unit Transformer LM + unit vocoder	2024.11
LLaMA-Omni	System	△	-	-	-	✓	-	-	-	✓	Whisper + units	Whisper encoder + LLaMA + streaming unit decoder	2024.09
Mini-Omni	System	△	-	-	-	✓	-	-	-	✓	SNAC codec	Whisper encoder + Qwen + SNAC token decoder	2024.08
GLM-4-Voice	System	✓	-	-	-	✓	-	-	-	✓	Speech tokenizer	Speech tokenizer + GLM-4 + flow-matching decoder	2024.12
Triadic VAP	Component	✓	-	-	✓	-	-	-	-	-	Multi-party waveform	Causal Transformer triadic VAP predictor	2025.08
OpenOmni	Model	△	-	-	-	✓	✓	-	-	✓	Speech tokens	Pretrained speech encoder + LLM + lightweight speech decoder	2025.01
GOAT-SLM	Model	-	-	-	-	✓	-	-	-	✓	Speech tokens	Dual semantic/acoustic heads + speech decoder	2025.07
OpenS2S	Model	-	-	-	-	✓	-	-	-	✓	Speech tokens	BLSP-Emo + streaming interleaved speech decoder	2025.07
Step-Audio 2	System	✓	-	-	-	✓	-	-	-	✓	Speech-text tokens	Step-Audio tokenizer + LLM + speech-token decoder	2025.07
InteractiveOmni	Model	△	-	-	-	✓	✓	-	-	✓	Audio-visual tokens	Audio-visual encoders + unified Transformer decoder	2025.10
Step-Audio	System	✓	-	-	-	✓	-	-	-	✓	Speech-text tokens	Step-Audio tokenizer + 130B LLM + speech decoder	2025.02
Baichuan-Audio	System	✓	-	-	-	✓	-	-	-	✓	Multi-codebook	Speech encoder + Baichuan LLM + multi-codebook decoder	2025.02
Kimi-Audio	Model	✓	-	-	-	✓	-	-	-	✓	Continuous + tokens	Audio encoder/tokenizer + Kimi LLM + speech decoder	2025.04
Qwen3-Omni	Model	✓	-	-	-	✓	✓	-	-	✓	Continuous + codec	Qwen Thinker + autoregressive Talker decoder	2025.09
PACHAT	System	△	-	-	✓	✓	-	-	✓	-	Speech frontend + TTS	Speech frontend + persona-aware LLM + expressive TTS	2025.11
VITA-Audio	Model	✓	-	-	-	✓	-	-	-	✓	Interleaved tokens	Audio encoder + LLM + interleaved speech-token decoder	2025.05
AV-Dialog	Model	-	-	-	-	✓	✓	-	-	✓	Audio-visual features	Audio-visual encoders + dialogue Transformer + speech decoder	2025.11
Qwen2.5-Omni	Model	✓	-	-	-	✓	✓	-	-	✓	Continuous + codec	Qwen Thinker + speech-token Talker + vocoder	2025.03
LLaMA-Omni 2	System	✓	-	-	-	✓	-	-	-	✓	Discrete units	Audio encoder + LLaMA + streaming unit decoder	2025.05
VoxMind	System	△	-	-	-	✓	-	-	-	✓	Speech tokens	Speech tokenizer + agentic dialogue LM + speech decoder	2026.07
ZipVoice-Dialog	Model	△	-	-	-	✓	-	-	-	✓	Continuous latent	Flow-matching DiT speech generator	2026.07
Interaction Mode: Half-duplex / Controlled Barge-in													
Incremental TT Model	Component	✓	-	-	-	-	-	-	-	-	Incremental features	Incremental neural TT predictor (no generator)	2019.09
Listener-aware BC	Component	△	-	-	✓	-	-	-	-	-	ASR text	Listener-aware neural BC predictor (no generator)	2020.05
BPM_MT	Component	△	-	-	✓	-	-	-	-	-	Speech + text	Multitask neural BC predictor (no generator)	2021.11
Device Directedness	Component	△	-	-	-	-	-	-	-	-	Acoustic + context	Acoustic-context directedness classifier	2022.11
Acoustic Barge-in	Component	△	✓	-	-	-	-	-	-	-	Acoustic + context	Contextual acoustic barge-in classifier	2022.09
Freeze-Omni	System	✓	✓	△	△	✓	-	-	-	✓	Continuous + units	Speech encoder + frozen LLM + chunkwise unit decoder	2024.11
Mini-Omni2	System	✓	✓	△	△	✓	✓	-	-	✓	SNAC codec	Whisper encoder + Qwen + SNAC token decoder	2024.10
VITA	System	✓	✓	△	△	✓	✓	-	✓	-	Encoder + TTS	Audio encoder + LLM + streaming TTS	2024.08
SHANKS	Add-on	✓	△	△	-	✓	-	-	✓	-	Streaming features	Streaming speech encoder + side-path state predictor	2025.10
Interaction Mode: Full-duplex													
Duplex Conversation (Outbound)	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / TTS	Streaming ASR + dialogue controller + TTS	2021.08
Duplex Conversation	System	✓	✓	✓	✓	✓	-	-	✓	-	ASR / TTS	Two-channel interaction model + ASR/TTS	2022.05
Synchronous LLM	Model	✓	✓	✓	✓	✓	-	-	-	✓	Speech units	Dual-channel speech units + synchronous LLaMA	2024.09
Moshi	Model	✓	✓	✓	✓	✓	-	-	-	✓	Mimi codec	Mimi codec + Helium temporal/depth Transformers	2024.10
SALMONN-omni	Model	✓	✓	✓	✓	✓	-	-	-	✓	Continuous embeddings	Speech encoder + codec-free asynchronous speech decoder	2024.11
Full-duplex LLM Scheme	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / LLM / TTS	Streaming ASR + LLM dialogue manager + TTS	2024.05
MinMo	Model	✓	✓	✓	✓	✓	✓	-	-	✓	Speech tokens	Speech encoder/tokenizer + 8B LLM + speech decoder	2025.01
SALM-Duplex	Model	✓	✓	✓	△	✓	-	-	-	✓	Encoder + codec	Streaming speech encoder + LLM + codec decoder	2025.05
OmniFlatten	Model	✓	✓	✓	✓	✓	-	-	-	✓	Speech-text tokens	Flattened speech-text streams + decoder-only Transformer	2025.07
Japanese FD System	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / LLM / TTS	Streaming ASR + LLM dialogue manager + TTS	2025.08
FireRedChat	System	✓	✓	✓	△	✓	-	✓	✓	-	ASR / TTS or codec	ASR/TTS cascade or codec-based semi-cascade	2025.09
FD Dialogue + DST	System	✓	✓	✓	△	✓	-	✓	✓	-	ASR / DST / TTS	Streaming ASR + DST/dialogue manager + TTS	2025.12
Chronological Thinking	Model	✓	✓	✓	✓	✓	-	-	-	✓	Synchronized tokens	Synchronized speech tokens + chronological Transformer	2025.10
LLM-Enhanced FD-DM	System	✓	✓	✓	△	✓	-	✓	-	-	ASR / LLM-DM / TTS	Streaming ASR + LLM-enhanced dialogue manager + TTS	2025.02
LSLM	Model	✓	✓	✓	△	✓	-	-	-	✓	SSL + speech tokens	SSL speech encoder + channel-fusion LLM + token decoder	2025.02
Chroma 1.0	System	✓	✓	✓	△	✓	-	-	-	✓	Speech tokens	Neural codec + real-time dialogue Transformer	2026.01
PersonaPlex	Model	✓	✓	✓	△	✓	-	-	-	✓	Neural codec	Mimi codec + duplex Transformer + voice conditioning	2026.02
MiniCPM-o 4.5	Model	✓	✓	✓	△	✓	✓	-	-	✓	Omni-modal stream	Omni encoder + MiniCPM LLM + streaming speech decoder	2026.04
DuplexSLA	Model	✓	✓	✓	✓	✓	-	-	-	✓	Continuous + discrete	Speech encoder + joint language/action LM + codec decoder	2026.05
BayLing-Duplex	Model	✓	✓	✓	✓	✓	-	-	-	✓	Audio tokens	Audio tokenizer + autoregressive duplex LLM + decoder	2026.06
F-Actor	Add-on	✓	✓	✓	✓	✓	-	-	✓	-	Control states	Behavior-state controller (uses host speech model)	2026.07
Hierarchical A-S Modeling	Model	✓	✓	✓	✓	✓	-	-	-	✓	Semantic + acoustic	Hierarchical semantic/acoustic duplex Transformer	2026.07
JoyAI-Talker	Model	✓	✓	✓	△	✓	-	-	-	✓	Semantic + acoustic	Semantic/acoustic tokenizers + dual-level Transformer	2026.08
SoulX-Duplug	Add-on	✓	✓	✓	△	✓	-	-	✓	-	Streaming ASR	Streaming ASR + semantic-state predictor (host TTS)	2026.03

Abbreviations: ASR, automatic speech recognition; TTS, text-to-speech; SSL, self-supervised learning; LLM, large language model; VAD, voice activity detection. Acoustic / Speech Model reports the principal speech encoder, tokenizer, dialogue backbone, decoder, or controller evaluated by each paper; Release uses YYYY.MM and follows the cited archival publication month, or the first arXiv month when the cited work is a preprint.

Development Trajectory: From Spoken Dialogue Models to Voice Agents

How to read the trajectory. Duplex task systems provide engineering context, while dGSLM (March 2022) marks the opening of the generative spoken-dialogue-model track. The table aligns the same papers used in Figure 1 along architecture/representation, interaction taxonomy, and agency/evaluation, making the transition from model-centric conversation to task-capable voice agents explicit rather than implied by chronology alone.

Phase	Period	Representative works	Architecture / Representation	Interaction Taxonomy	Agency / Evaluation Frontier
Engineering precursor	2021.09	Duplex Conversation (Outbound)	Cascaded ASR-dialogue-policy-TTS stack with an explicit interruption and control layer.	Controller-mediated duplex / barge-in; listening and speaking states are externally orchestrated.	Domain-bound task completion; no unified generative spoken-dialogue model.
Generative model-track opening	2022.03-05	dGSLM; Duplex Conversation	Textless dual-channel speech generation and learned conversational timing.	Turn-taking plus learned overlap and backchannel timing; duplex becomes a modeling target.	Open-domain spoken generation emerges; instruction following and tool use remain absent.
Speech-native instruction foundation	2023.05-06	SpeechGPT; AudioPaLM	Discrete speech units, LLM backbones, speech-text alignment, and cross-modal instruction tuning.	Predominantly turn-based end-to-end speech input/output.	Spoken instruction following and multimodal transfer improve, but real-time synchrony is limited.
Real-time and duplex diversification	2024.08-11	VITA; Mini-Omni; Synchronous LLMs; Moshi; Mini-Omni2; OmniFlatten; SALMONN-omni	Streaming/interleaved token generation, parallel semantic-acoustic streams, and dual-stream decoder variants.	Streaming barge-in expands toward full-duplex overlap, interruption, and backchannels.	Conversation becomes more fluid; external action and persistent planning are not yet central.
Scaled unified voice models	2025.01-09	MinMo; Step-Audio; Baichuan-Audio; Qwen2.5-Omni; VITA-Audio; SALM-Duplex; FireRedChat	Unified omni-modal LLMs, interleaved audio/text tokens, plus end-to-end and semi-cascaded variants.	Low-latency streaming, controlled barge-in, and increasingly robust duplex operation.	Instruction following and robustness scale up; tool orchestration is still mostly external.
Tool-augmented voice agents	2025.10	Stream RAG; VoiceAgentBench	Streaming retrieval and tool interfaces are layered onto spoken-dialogue backbones.	Tool calls can begin during incoming speech without ending the conversational stream.	Multi-tool, multi-turn task completion becomes explicit and benchmarked.
Controllable full-duplex agents	2026.02-04	PersonaPlex; SoulX-Duplug; MiniCPM-o 4.5; Full-Duplex-Bench-v3	Persona conditioning, streaming state prediction, and real-time omni-modal generation converge.	Dynamic turn state, interruption handling, overlap, and controllable full-duplex behavior.	Role control and realistic tool-use evaluation expose reliability gaps under disfluency.
Reasoning and action integration	2026.05-07	DuplexSLA; VoxMind; Dual-Reasoner	Synchronized speech-language-action generation and thinking-while-talking reasoning.	Concurrent speech, listening, reasoning, and action generation close the interaction loop.	Reason-plan-act behavior is integrated with speech; memory, safety, and long-horizon autonomy remain open.

Trajectory Synthesis

- Representation: cascaded ASR/TTS -> textless dual-channel units -> speech-native LLMs -> parallel semantic/acoustic streams -> synchronized speech-language-action generation.
- Interaction: controller-mediated barge-in -> learned turn timing -> streaming interruption -> full-duplex overlap/backchannels -> concurrent speech and action.
- Agency: domain task logic -> generative dialogue -> spoken instruction following -> retrieval/tool use -> reason-plan-act behavior and persona control.
- Evidence: speech quality and timing -> latency/interruption -> overlap naturalness -> tool correctness and task completion -> memory, safety, and agent reliability.

Scope note. Every representative work in the trajectory is already listed in the preceding collection; the table reorganizes existing evidence and introduces no bibliography entry solely for this synthesis.

Development Roadmap and Capability Curve

The main curve summarizes a qualitative progression from generative spoken-dialogue modeling to integrated spoken agents. Year branches expand representative works from this collection: color denotes the interaction or agency path, while dashed tags identify benchmark or training evidence.

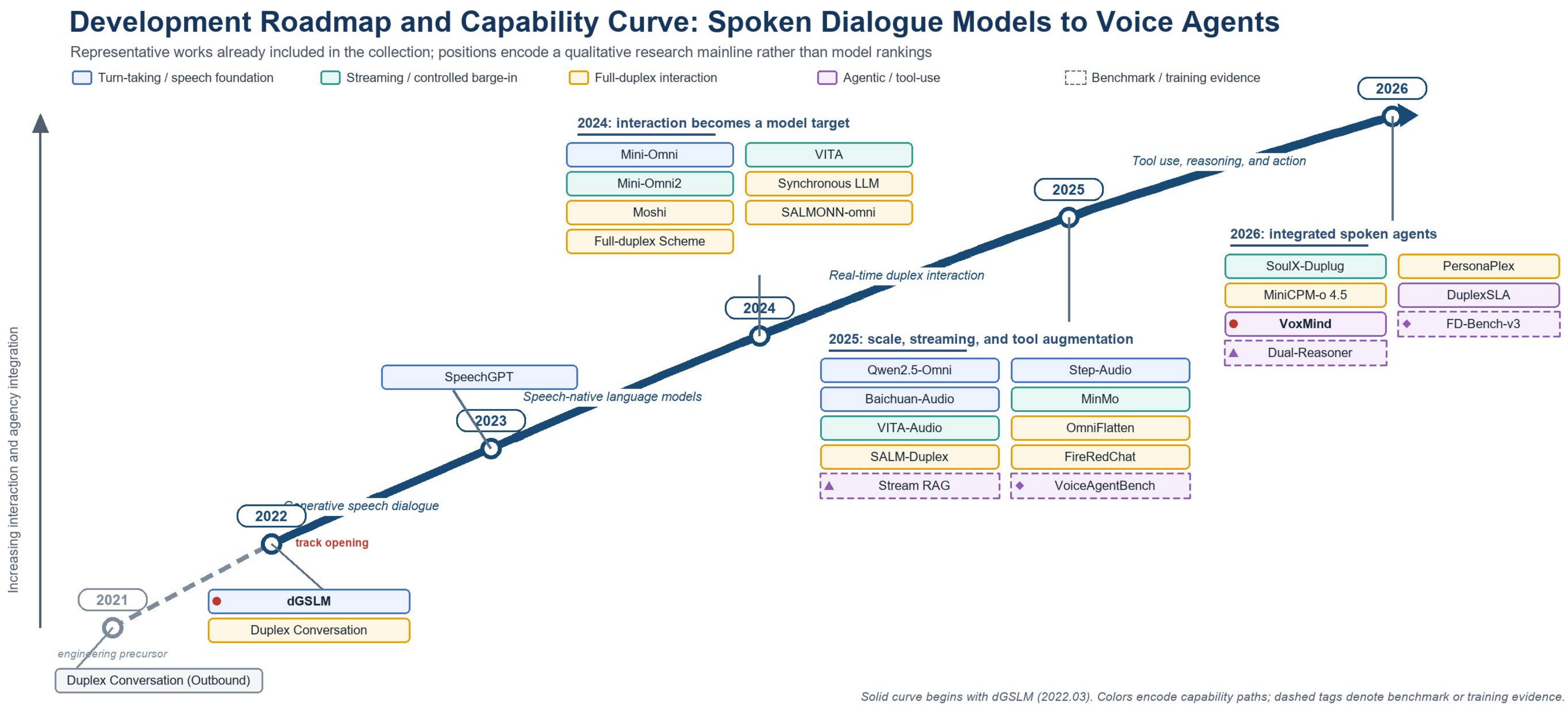

Figure 1. Academic roadmap and conceptual capability curve for Multi-Turn Speech Dialogue. The dotted 2021 segment provides engineering context; the solid model-development mainline begins with dGSLM in March 2022. Branches show representative works already included in this collection.

Interpretation. The field evolves along three coupled axes: speech-native representation, increasingly synchronous interaction, and growing task agency. The dense expansion after 2024 reflects diversification rather than a single winning architecture; by 2025-2026, retrieval, tool use, reasoning, action generation, and agent-oriented evaluation begin to close the loop from model-centric conversation to task-capable voice agents.